

4. Heating is a common method used to preserve food.

- (a) **State three** methods, **other than heating**, in which the growth of bacteria can be slowed down or stopped in food products.

Explain how each method works.

[6 QER]

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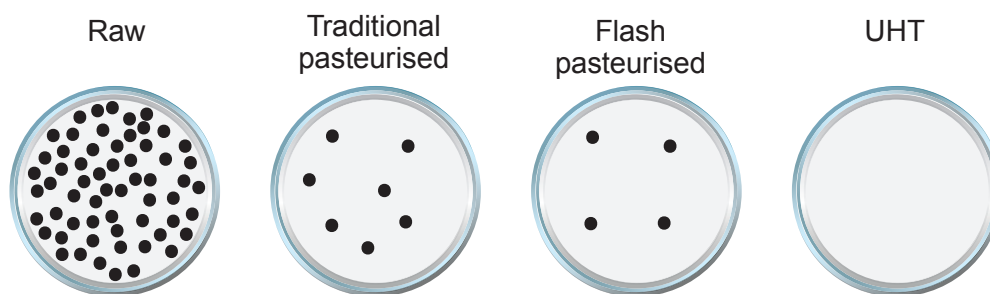
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- (b) Scientists investigated four types of milk. They monitored the growth of bacterial colonies on agar plates using 0.1 cm^3 from each type of milk over two days. Their results are shown below.



Type of milk	Heat treatment	Number of bacterial colonies from 0.1 cm^3 of milk	Number of bacterial colonies in a 40 cm^3 serving
raw	none	60	24 000
traditional pasteurised	63°C for 30 minutes	7	
flash pasteurised	78°C for 35 seconds		1 600
ultra-high temperature pasteurised (UHT)	135°C for 2 seconds	none	none

- (i) Complete the table.

[2]

Space for working

- (ii) It is claimed that unopened UHT milk can be stored on supermarket shelves at room temperature for six months. Explain whether you agree with the claim. [2]

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